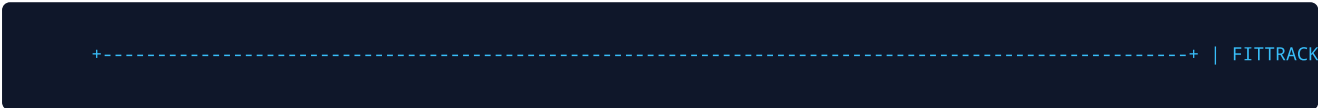


Strict Baseline Scope Boundary: Architectural specifications govern the baseline implementation only (Authentication, Workout CRUD, Search, and Google Gemini AI integration). Out-of-scope future extensions (Nutrition Tracking, BMI Calculation, Wearable Integration, Workout Reminders, and Analytics Dashboards) are strictly isolated from the baseline runtime.

1. LAYERED MVC ARCHITECTURAL BLUEPRINT

FitTrack AI utilizes a layered Model-View-Controller (MVC) architectural pattern decoupling request routing, security inspection, business logic, data modeling, and external AI services.



2. ARCHITECTURAL PRINCIPLES & TOPOLOGY

SEPARATION OF CONCERNS Decouples routing declarations, controller logic, Mongoose persistence schemas, and AI integration into dedicated, testable directories.	STATELESS IDENTITY Eliminates server session state by validating digitally signed JSON Web Tokens (JWT) on every protected transaction.	CRYPTOGRAPHIC ISOLATION Enforces multi-tenant record isolation by scoping all workout queries and mutations strictly to the authenticated user ID (<code>req.user._id</code>).
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3. SYSTEM COMPONENTS & OPERATIONAL RESPONSIBILITIES

COMPONENT LAYER	MODULES & FILES	ARCHITECTURAL RESPONSIBILITIES	DOCUMENTED TECHNOLOGIES
Server Gateway	server.js config/db.js	Bootstraps the Express application, loads environment variables via dotenv, configures HTTP port binding, and initializes Mongoose connection to MongoDB.	Node.js (v16+), Express.js, dotenv, Mongoose
Middleware Shield	middleware/auth.js middleware/error.js	Inspects cross-origin permissions, parses incoming JSON payloads, extracts and verifies Bearer JWT tokens, injects req.user, and standardizes application exceptions.	CORS, jsonwebtoken, Express error handler
Routing Dispatcher	routes/authRoutes.js routes/workoutRoutes.js routes/aiRoutes.js	Defines explicit REST endpoints, binds appropriate HTTP verbs (GET, POST, PUT, DELETE), and mounts authentication guards on protected endpoints.	Express.Router
Business Controllers	controllers/authController.js controllers/workoutController.js controllers/aiController.js	Orchestrates application logic: validates inputs, enforces record ownership, coordinates password hashing, signs tokens, aggregates workouts, and handles AI requests.	JavaScript (ES6+ / Node.js)
Data Access Models	models/User.js models/Workout.js	Defines schema constraints, data types, required rules, unique indexes, and foreign key relationships (Workout.userId referencing User._id).	Mongoose, MongoDB
Service & Integration	bcryptjs, jsonwebtoken, Google Gemini API Client	Executes one-way cryptographic hashing, signs/verifies JWT bearer credentials, and issues external HTTPS generative inference calls using GEMINI_API_KEY.	bcryptjs, jsonwebtoken, Google Gemini API

4. END-TO-END PROCESSING FLOWS

AUTHENTICATION FLOW (REGISTRATION / LOGIN)

- Client dispatches POST /api/auth/register or /api/auth/login with credentials.
- Gateway applies CORS and body-parser middleware.
- authController checks MongoDB for email presence.
- Registration salts and hashes password via bcryptjs; login verifies hash using bcrypt.compare().
- System generates signed JWT with JWT_SECRET and returns HTTP 200/201.

PROTECTED WORKOUT & SEARCH FLOW

- Client sends request with Authorization: Bearer <token>.
- authMiddleware verifies JWT; injects user context into req.user.
- workoutController applies CRUD logic or builds multi-criteria search filter (name, category, date).
- Mongoose validates schema and executes query enforcing user ID scope.
- Returns isolated JSON workout payload with HTTP 200/201.

GOOGLE GEMINI AI REGIMEN RECOMMENDATION FLOW

- Authenticated user sends POST /api/ai/recommend with goal and fitnessLevel.
- Auth Guard validates token; aiController validates input presence.
- Controller constructs structured prompt contextualizing user fitness goals.
- Dispatches authorized HTTPS request to Google Gemini API via GEMINI_API_KEY.
- Gemini generative response returned to client as standardized JSON payload.

AI HISTORICAL FITNESS INSIGHTS FLOW

- Authenticated user requests GET /api/ai/insights.
- Auth Guard validates token; aiController retrieves user's historical workout documents from MongoDB.
- Controller aggregates logs into structured analytical prompt.
- Dispatches prompt to Google Gemini API to analyze consistency and progression.
- Returns generative workout analysis and actionable recommendations.

5. ARCHITECTURE-TO-IMPLEMENTATION TRACEABILITY MATRIX

Exhaustive mapping connecting architectural layers, HTTP methods, endpoints, controllers, models, and security middleware shields.

ARCHITECTURAL MODULE	HTTP VERB	TARGET ENDPOINT	CONTROLLER HANDLER	DATA MODEL	SECURITY SHIELD
Authentication	POST	/api/auth/register	authController.register	User	bcryptjs Hash / CORS
Authentication	POST	/api/auth/login	authController.login	User	bcryptjs Verify / JWT Sign
Authentication	GET	/api/auth/profile	authController.getProfile	User	Bearer JWT Guard
Workout Management	POST	/api/workouts	workoutController.createWorkout	Workout	Bearer JWT / Schema Validation
Workout Management	GET	/api/workouts	workoutController.getWorkouts	Workout	Bearer JWT / Tenant Scoping
Workout Management	GET	/api/workouts/:id	workoutController.getWorkoutById	Workout	Bearer JWT / Tenant Isolation
Workout Management	PUT	/api/workouts/:id	workoutController.updateWorkout	Workout	Bearer JWT / Ownership Verification
Workout Management	DELETE	/api/workouts/:id	workoutController.deleteWorkout	Workout	Bearer JWT / Ownership Verification
Workout Search	GET	/api/workouts?category=..	workoutController.getWorkouts	Workout	Bearer JWT / Query Sanitization
AI Intelligence	POST	/api/ai/recommend	aiController.getRecommendation	N/A (External)	Bearer JWT / GEMINI_API_KEY
AI Intelligence	GET	/api/ai/insights	aiController.getInsights	Workout	Bearer JWT / Aggregation Scoping
System Exception	ALL	Global Application Scope	middleware/error.js	All Models	Sanitized Error DTO / 500 Guard

6. ARCHITECTURAL BOUNDARY ENFORCEMENT

Strict baseline containment isolating core services from documented out-of-scope future capabilities.

FEATURE CAPABILITY	ARCHITECTURAL SCOPE	BOUNDARY ENFORCEMENT & ISOLATION STRATEGY
Auth, CRUD, Search & AI	BASELINE IMPLEMENTATION	Active components deployed across current MVC layers, routes, and data models.
Nutrition Tracking	FUTURE EXTENSION	Strictly excluded from baseline Mongoose models, routing declarations, and controllers.
BMI Calculation	FUTURE EXTENSION	Strictly excluded; no biometrics or anthropometric fields defined in baseline User schema.
Wearable Integration	FUTURE EXTENSION	Strictly excluded; no IoT, Bluetooth, or external health syncing interfaces implemented.
Workout Reminders	FUTURE EXTENSION	Strictly excluded; no scheduler, cron jobs, or push notification brokers included in baseline.
Analytics Dashboards	FUTURE EXTENSION	Strictly excluded; baseline limited to REST JSON responses; UI dashboards reserved for future.